

Machine Learning

Instructor: Hafiz Abdul Rehman

Lecture 06: Performance Measures

The Output of a Classifier

#	Height (inches)	Weight (kgs)	B.P. Sys	B.P. Dia	Heart disease	
	x				y	$h(x)$
1	62	70	120	80	No	No
2	72	90	110	70	No	Yes
3	74	80	130	70	No	No
4	65	120	150	90	Yes	Yes
5	67	100	140	85	Yes	No
6	64	110	130	90	No	Yes
7	69	150	170	100	Yes	Yes
8	75	127	160	95	Yes	No
9	66	66	135	90	Yes	Yes

- y : Gold standards / Gold labels / Ground truth
- $h(x)$: Predicted labels

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Negative

Positive

Negative

Positive

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 - Positives and Negatives

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True	Negative
False	Positive
True	Negative
True	Positive
False	Negative
False	Positive
True	Positive
False	Negative
True	Positive

- y : Gold standards / Gold labels / Ground truth
- $h(x)$: Predicted labels
 - Positives and Negatives
 - True (matching the ground truth) or False (not matching the ground truth)

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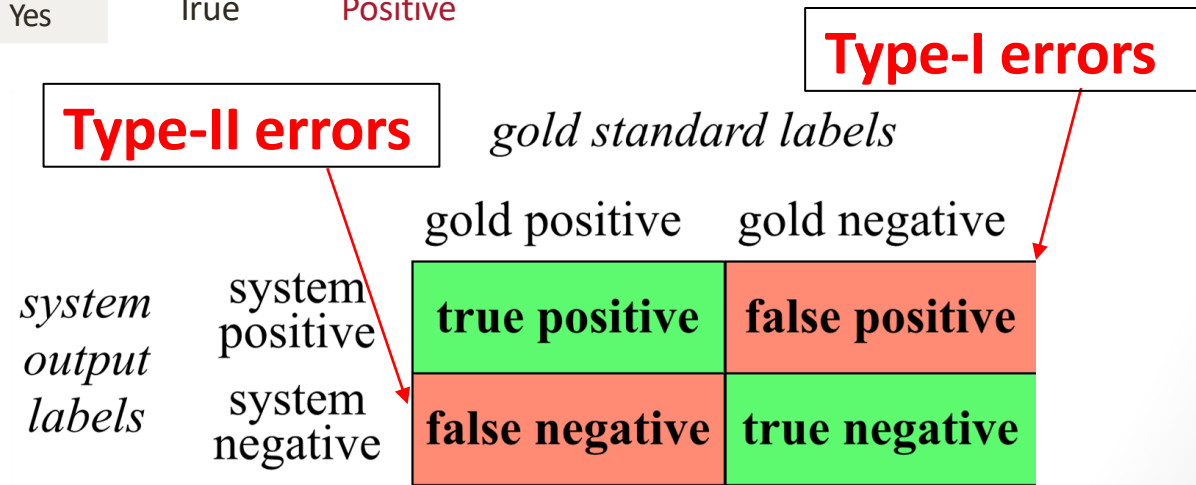
True	Negative	
False	Positive	Type-I Error
True	Negative	
True	Positive	
False	Negative	Type-II Error
False	Positive	Type-I Error
True	Positive	
False	Negative	Type-II Error
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Confusion Matrix

		Gold Labels	
		Gold Positive	Gold Negative
Predicted Labels	Predicted Positive	True Positives (tp)	False Positives (fp)
	Predicted Negative	False Negatives (fn)	True Negatives (tn)

Confusion Matrix

$$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$$

		Gold Labels	
		Gold Positive	Gold Negative
Predicted Labels	Predicted Positive	True Positives (<i>tp</i>)	False Positives (<i>fp</i>)
	Predicted Negative	False Negatives (<i>fn</i>)	True Negatives (<i>tn</i>)

- You are the CEO of a Pie Company and you need to know what people are saying about your pies on social media
- You build a system that detects tweets concerning your pies
 - the positive class is tweets about your pies
 - the negative class is all other tweets
- Imagine that we looked at a million tweets
 - 100 of them are discussing their love (or hatred) for our pie
 - the other 999,900, are tweets about other matters
- You created a classifier that stupidly classified every tweet as “not about pie”
 - 999,900 true negatives, 100 false negatives, 0 false positives, and 0 true positives, for an accuracy of 999,900/1,000,000 or 99.99%!
- **Accuracy is not a good metric when the goal is to discover a rare event, or at least not completely balanced in frequency**
- **A very common situation in the world!**

Confusion Matrix

		Gold Labels			
		Gold Positive	Gold Negative		
Predicted Labels	Predicted Positive	True Positives (tp)	False Positives (fp)	$\frac{tp}{tp + fp}$	"Precision" aka "Positive Predictive Value"
	Predicted Negative	False Negatives (fn)	True Negatives (tn)	$\frac{tn}{fn + tn}$	"Negative Predictive Value"
		$\frac{tp}{tp + fn}$ "Recall" aka "Sensitivity" aka "True Positive Rate"	$\frac{tn}{fp + tn}$ "Specificity" aka "True Negative Rate"	$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$	

Performance Metrics

$$\text{Accuracy} = \frac{\text{My Correct Answers}}{\text{All Questions}} = \frac{tp + tn}{tp + tn + fp + fn}$$

- What fraction of time am I correct in my classification

$$\text{Precision} = \frac{\text{True Positives}}{\text{My Positives}} = \frac{tp}{tp + fp}$$

- How much should you trust me when I say that something tests positive
- What fraction of my positives are true positives

$$\text{Recall} = \text{Sensitivity} = \frac{\text{True Positives}}{\text{Real Positives}} = \frac{tp}{tp + fn}$$

- How much of the reality has been covered by my positive output?
- What fraction of the true positives is captured by my positives? E.g. How many sick people are correctly identified as having the condition?

$$\text{Specificity} = \frac{\text{True Negatives}}{\text{Real Negatives}} = \frac{tn}{tn + fp}$$

- How much of the reality has been covered by my negative output?
- What fraction of the true negatives is captured by my negatives? E.g. How many identified healthy people do not have the condition?

A worked example

You are shown a set of 21 coins: 10 gold and 11 copper. Your task to accept all gold coins and reject all copper ones

You accept 7 coins as being gold (these are your positives)

- 5 of these are actually gold (these are your true positives, tp)
- 2 of these are copper (these are your false positives, fp)
- You falsely rejected 5 gold ones (false negatives, fn)
- You correctly rejected 9 copper ones (true negatives, tn)

	Actual Gold	Actual Copper
Predicted Gold	5	2
Predicted Copper	5	9

- Your **precision** is $\frac{tp}{\text{all of your positives}} = \frac{tp}{tp+fp} = \frac{5}{7}$
- Your **recall** is $\frac{tp}{\text{number of actual gold coins}} = \frac{tp}{tp+fn} = \frac{5}{10}$
- Your **specificity** is $\frac{tn}{\text{number of copper coins}} = \frac{tn}{fn+tn} = \frac{9}{11}$
- Your **accuracy** is $\frac{\text{correct answers}}{\text{all attempts}} = \frac{tp+tn}{tp+tn+fp+fn} = \frac{5+9}{5+9+2+5} = \frac{14}{21}$



Realistic extremes

- You accept only one coin and that is gold
 - Your precision is very high (1/1) but recall is very low (1/10)
- You return all 21 coins
 - Your recall is very high (10/10) but precision is very low (10/21)
- Only one out of the 21 coins is gold. And you reject
- everything.
 - Your accuracy is very high ($20/21 = 0.95$) but precision/recall are 0
- So, what do we do now?
- A combined measure?

	Ac Gld	Ac Cop
Pr Gld	1	0
Pr Cop	9	11

	Ac Gld	Ac Cop
Pr Gld	10	11
Pr Cop	0	0

	Ac Gld	Ac Cop
Pr Gld	0	0
Pr Cop	1	20

Confusion Matrix

		Gold Labels			
		Gold Positive	Gold Negative		
Predicted Labels	Predicted Positive	True Positives (tp)	False Positives (fp)	$\frac{tp}{tp + fp}$	"Precision" aka "Positive Predictive Value"
	Predicted Negative	False Negatives (fn)	True Negatives (tn)	$\frac{tn}{fn + tn}$	"Negative Predictive Value"
		$\frac{tp}{tp + fn}$ "Recall" aka "Sensitivity" aka "True Positive Rate"	$\frac{tn}{fp + tn}$ "Specificity" aka "True Negative Rate"	$Accuracy = \frac{tp + tn}{tp + fp + tn + fn}$	
		$\frac{fn}{tp + fn}$ 1 - Sensitivity = "False Negative Rate" aka "False Rejection Rate"	$\frac{fp}{fp + tn}$ 1 - Specificity = "False Positive Rate" aka "False Acceptance Rate"	<u>True Positive Rate</u> = "Positive Likelihood Ratio" <u>False Positive Rate</u> <u>False Negative Rate</u> = "Negative Likelihood Ratio" True <u>Negative Rate</u>	

F1 Measure

- It is useful to have a single number to describe performance. Should be high when both P and R are high.
- A combined measure that assesses the P/R tradeoff is F measure (weighted harmonic mean of precision and recall):

$$F = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2PR}{P + R}$$

F_β Measure

- We can choose to favor precision or recall by using an interpolation weight α :

$$F = \frac{2}{\alpha \frac{1}{P} + (1 - \alpha) \frac{1}{R}}$$
$$= \frac{(\beta^2 + 1)RP}{\beta^2 P + R} \qquad \beta = \frac{1 - \alpha}{\alpha}$$

- Balanced F1 measure has $\beta = 1$ (that is, $\alpha = 1/2$) as shown above.
- To give more weight to the Precision, we pick a β value in the interval
 - $0 < \beta < 1$.
 - notice that it is getting multiplied with P in the denominator
- To give more weight to the Recall, we pick a β Value in the interval
 - $1 < \beta < +\infty$
- $\beta \rightarrow 0$ considers only precision, $\beta \rightarrow +\infty$ only recall

F_β Measure

$$F_\beta = \frac{(\beta^2 + 1)RP}{\beta^2 P + R}$$

- The β parameter differentially weights the importance of recall and precision
 - based on the needs of an application
 - values of $\beta > 1$ favor recall, while
 - values of $\beta < 1$ favor precision
- When $\beta = 1$, precision and recall are equally balanced
- This is the most frequently used metric, and is called $F_{\beta=1}$ or just F1:

$$F1 = \frac{2PR}{P + R}$$

Thank You